

Final Capstone Project

Capstone Project: Enterprise Vendor Invoice Management System

Project Overview

Build a production-ready **Vendor Invoice Management System** that demonstrates mastery of SAP CAP, HANA Cloud, Fiori Elements, S/4HANA integration, security, and cloud deployment.

Business Scenario

Your organization processes 500+ vendor invoices monthly across multiple departments. The current manual process causes delays, approval bottlenecks, and compliance issues. You must build an automated system that:

- Centralizes vendor and invoice data
- Enforces approval workflows with role-based access
- Integrates with S/4HANA for vendor master sync
- Provides real-time status tracking and analytics
- Accessible via SAP Build Work Zone for all stakeholders

Learning Objectives

By completing this project, you will demonstrate:

Data Modeling - Complex CDS models with associations, compositions, and aspects

Service Layer - Custom handlers, actions, validations, and business logic **UI**

Development - Fiori Elements with advanced annotations and navigation

Security - Authentication, authorization, and row-level security

Integration - S/4HANA OData API consumption

Deployment - MTA build, Cloud Foundry deployment, HANA Cloud

User Experience - Build Work Zone configuration for centralized access

Functional Requirements

Core Features

Feature	Description
Vendor Master	Maintain vendor records with approval status
Invoice Management	Create, read, update, delete invoices with draft support
Approval Workflow	Submit → Pending → Approved/Rejected states Role-
Based Access	Admin, Vendor Manager, Approver, Viewer roles Status
Tracking	Real-time invoice status monitoring
S/4HANA Sync	Import vendor master data from S/4HANA Reporting
	Invoice analytics by status, vendor, and date range
Work Zone Access	Single point of access for all users

Business Rules

1. Only vendors with **status = 'APPROVED'** can have invoices created
 2. Invoice amount must be > 0 and $\leq 1,000,000$
 3. Invoice date cannot be future dated
 4. Only invoices in **SUBMITTED** status can be approved/rejected
 5. Approver cannot approve their own submitted invoices
 6. Vendor Managers can only view invoices for their assigned vendors
 7. Invoice number must be unique per vendor
 8. Line items must sum up to the invoice total amount
 9. At least one line item is mandatory for invoice submission
 10. Deleted vendors cannot have new invoices created
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Detailed Logic Specifications

Invoice Status State Machine

None

DRAFT → SUBMITTED → APPROVED → PAID



REJECTED

Status Transitions:

- **DRAFT → SUBMITTED:** Triggered by "Submit for Approval" action
- **SUBMITTED → APPROVED:** Triggered by "Approve Invoice" action
- **SUBMITTED → REJECTED:** Triggered by "Reject Invoice" action
- **APPROVED → PAID:** (Optional bonus - triggered by "Mark as Paid" action)

Invalid Transitions (Must Show Error):

- DRAFT → APPROVED (cannot skip submission)
- APPROVED → REJECTED (cannot reject after approval)
- REJECTED → APPROVED (cannot approve after rejection)
- SUBMITTED → DRAFT (cannot revert to draft after submission)

Action Specifications

Action 1: Create New Invoice

Trigger: User clicks "Create" button on Invoice list page

Preconditions Check:

- User must have Admin or VendorManager role
- If VendorManager: Must have at least one assigned vendor

System Behavior:

1. Open invoice creation form in draft mode
2. Set status = 'DRAFT' (auto-populated, read-only)
3. Set createdBy = current logged-in user (auto-populated)
4. Set createdAt = current timestamp (auto-populated)
5. Enable vendor dropdown (show only APPROVED vendors)
6. Set invoice date = today's date (default, editable)
7. Set due date = invoice date + 30 days (default, editable)
8. Currency = default from selected vendor (auto-populated after vendor selection)

Validation on Save:

- Vendor: MANDATORY, must be from APPROVED vendors only
 - Error message: "Please select an approved vendor"
- Invoice Number: MANDATORY, must be unique for the selected vendor
 - Error message: "Invoice number [NUMBER] already exists for vendor [VENDOR_NAME]"
- Invoice Date: MANDATORY, cannot be future date
 - Error message: "Invoice date cannot be in the future"
- Due Date: MANDATORY, must be $\geq$ invoice date
 - Error message: "Due date must be on or after invoice date"
- Amount: MANDATORY, must be > 0 and $\leq 1,000,000$
 - Error message: "Amount must be between 0.01 and 1,000,000"
- Currency: MANDATORY
 - Error message: "Currency is required"

Success Output:

- Invoice created with status = 'DRAFT'
 - Success message: "Invoice [INVOICE_NUMBER] created successfully"
 - Navigate to invoice detail page
 - Show "Submit for Approval" button (enabled)
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Action 2: Add Line Item to Invoice

Trigger: User clicks "Add" button in Line Items table (within invoice detail page)

Preconditions:

- Invoice must be in DRAFT status
- User must be the creator of the invoice or have Admin role

System Behavior:

1. Add new row to line items table
2. Line number = auto-increment from last line item (1, 2, 3...)
3. All fields editable

Required Fields per Line Item:

- Description: MANDATORY, max 255 characters
- Quantity: MANDATORY, must be > 0
- Unit Price: MANDATORY, must be > 0
- Total Amount: AUTO-CALCULATED = Quantity × Unit Price (read-only field)

Validation:

- Cannot save invoice without at least one line item for submission
- Description cannot be empty
 - Error: "Line item description is required"
- Quantity must be positive number
 - Error: "Quantity must be greater than zero"
- Unit price must be positive number
 - Error: "Unit price must be greater than zero"

On Save Draft:

- Calculate sum of all line items' total amounts
- Display sum below line items table
- Warning if sum ≠ invoice header amount: "Line items total ([SUM]) does not match invoice amount ([AMOUNT]). Please adjust before submission."

Action 3: Submit for Approval

Trigger: User clicks "Submit for Approval" button on invoice detail page

Preconditions Check (All must pass):

1. Invoice status = 'DRAFT'
 - Error: "Only draft invoices can be submitted"
2. At least one line item exists
 - Error: "Invoice must have at least one line item"
3. Sum of line items total = Invoice header amount
 - Error: "Line items total ([SUM]) does not match invoice amount ([AMOUNT]). Adjustment required."
4. All mandatory fields filled

- Error: "Please complete all required fields"
- 5. User must be creator or Admin
 - Error: "You do not have permission to submit this invoice"

System Behavior on Success:

1. Change status from 'DRAFT' to 'SUBMITTED'
2. Set submittedBy = current user ID
3. Set submittedAt = current timestamp
4. Create entry in ApprovalHistory:
 - Action = 'SUBMITTED'
 - Actor = current user name
 - Timestamp = current timestamp
 - Comments = "Invoice submitted for approval"
5. Disable all editing fields (invoice becomes read-only)
6. Hide "Submit for Approval" button
7. Show status badge as "SUBMITTED" with Yellow/Warning color
8. Display info message: "Invoice [INVOICE_NUMBER] submitted for approval successfully"
9. Send notification to Approver role users (if notification system implemented)

Post-Submission State:

- All fields become read-only
 - Line items cannot be added/edited/deleted
 - Attachments can still be added
 - Only "Approve" and "Reject" actions visible (for Approver role only)
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Action 4: Approve Invoice

Trigger: User clicks "Approve" button on submitted invoice detail page

Preconditions Check:

1. Invoice status = 'SUBMITTED'
 - Error: "Only submitted invoices can be approved"
2. User has Approver or Admin role
 - Error: "You do not have permission to approve invoices"
3. Approver ≠ Submitter (current user ID ≠ submittedBy user ID)
 - Error: "You cannot approve your own submitted invoice"

Input Required:

- Comments (optional text field, max 500 characters)

System Behavior on Success:

1. Change status from 'SUBMITTED' to 'APPROVED'

2. Set approvedBy = current user ID
3. Set approvedAt = current timestamp
4. Set approvalComments = entered comments (if provided)
5. Create entry in ApprovalHistory:
 - Action = 'APPROVED'
 - Actor = current user name
 - Timestamp = current timestamp
 - Comments = entered comments or "Invoice approved"
6. Show status badge as "APPROVED" with Green/Success color
7. Hide "Approve" and "Reject" buttons
8. Display success message: "Invoice [INVOICE_NUMBER] approved successfully by [APPROVER_NAME]"
9. Send notification to invoice creator (if notification system implemented)

UI Changes:

- Status indicator changes to green "APPROVED"
 - Show approval date and approver name in header
 - Show approval comments (if provided)
 - Enable "Mark as Paid" action (if bonus feature implemented)
 - Invoice remains read-only
-

Action 5: Reject Invoice

Trigger: User clicks "Reject" button on submitted invoice detail page

Preconditions Check:

1. Invoice status = 'SUBMITTED'
 - Error: "Only submitted invoices can be rejected"
2. User has Approver or Admin role
 - Error: "You do not have permission to reject invoices"

Input Required:

- Rejection Reason (MANDATORY text field, max 500 characters)
 - Cannot be empty
 - Error: "Rejection reason is mandatory"

System Behavior on Success:

1. Change status from 'SUBMITTED' to 'REJECTED'
2. Set rejectedBy = current user ID
3. Set rejectedAt = current timestamp
4. Set rejectionReason = entered reason
5. Create entry in ApprovalHistory:
 - Action = 'REJECTED'

- Actor = current user name
- Timestamp = current timestamp
- Comments = entered rejection reason
- 6. Show status badge as "REJECTED" with Red/Error color
- 7. Hide "Approve" and "Reject" buttons
- 8. Display rejection message: "Invoice [INVOICE_NUMBER] rejected by [REJECTOR_NAME]"
- 9. Send notification to invoice creator with rejection reason (if notification system implemented)

UI Changes:

- Status indicator changes to red "REJECTED"
 - Show rejection date and rejector name in header
 - Show rejection reason prominently (e.g., in alert box)
 - Invoice remains read-only
 - No further actions available on rejected invoice
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Action 6: Sync Vendors from S/4HANA

Trigger: User clicks "Sync Vendors from S/4HANA" tile/button in Work Zone or Admin panel

Preconditions:

- User must have Admin role
 - Error: "Only administrators can sync vendor data"
- S/4HANA destination must be configured
 - Error: "S/4HANA system connection not configured"

System Behavior:

1. Show loading indicator: "Syncing vendors from S/4HANA..."
2. Connect to S/4HANA Business Partner API
3. Fetch all suppliers from S/4HANA system
4. For each supplier:
 - Check if vendor exists in local database (match by external ID or tax ID)
 - If exists: UPDATE local vendor record with latest S/4HANA data
 - If not exists: CREATE new vendor record with status = 'PENDING'
5. Map S/4HANA fields to local fields:
 - Supplier Name → Vendor Name
 - Supplier Address → Vendor Address
 - Supplier Email → Vendor Email
 - Supplier Phone → Vendor Phone
 - Country → Country
 - Currency → Currency
 - Tax Number → Tax ID
 - S/4HANA Supplier ID → External System ID

Success Output:

- Display summary message:

None

Vendor sync completed successfully!

- Total vendors in S/4HANA: [COUNT]
- New vendors created: [COUNT]
- Existing vendors updated: [COUNT]
- Sync completed at: [TIMESTAMP]

- Refresh vendor list to show newly synced vendors
- New vendors appear with status = 'PENDING' (require approval before use)

Error Handling:

- If connection fails: "Failed to connect to S/4HANA system. Please check destination configuration."
- If partial sync: "Sync completed with errors. [SUCCESS_COUNT] vendors synced successfully. [FAIL_COUNT] vendors failed."
- Show detailed error log accessible to Admin

UI Behavior Specifications

Invoice List Page (Admin View)

Initial Load:

- Display all invoices in a table with columns:
 - Invoice Number (clickable link)
 - Vendor Name (with navigation to vendor)
 - Invoice Date (formatted as DD-MMM-YYYY)
 - Due Date (formatted as DD-MMM-YYYY)
 - Amount (formatted with currency symbol)
 - Currency Code
 - Status (with colored badge)
- Default sorting: Invoice Date descending (newest first)
- Pagination: 20 records per page

Status Color Coding:

- DRAFT: Grey/Neutral
- SUBMITTED: Yellow/Warning
- APPROVED: Green/Success
- REJECTED: Red/Error
- PAID: Blue/Info

Filter Bar (Above table):

- Vendor dropdown (multi-select, shows all vendors)
- Status dropdown (multi-select, shows all status values)
- Invoice Date Range (From Date - To Date picker)
- Amount Range (Min - Max input fields)
- "Go" button to apply filters
- "Clear" button to reset all filters

Search Bar:

- Free text search across invoice number, vendor name
- Search triggers on "Enter" key or search icon click
- Clear search icon appears when text entered

Actions:

- "Create" button (top right) → Opens invoice creation form
 - Click on invoice number → Opens invoice detail page
 - Export to Excel button → Downloads filtered data as Excel
-

Invoice Detail Page

Header Section:

- Invoice Number (large, prominent)
- Status Badge (colored, as per status)
- Vendor Name (clickable link to vendor detail)
- Created By user name and date
- If SUBMITTED: Show "Submitted by [USER] on [DATE]"
- If APPROVED: Show "Approved by [USER] on [DATE]"
- If REJECTED: Show "Rejected by [USER] on [DATE]" with reason in red alert box

General Information Section (Facet 1): Display fields in 2-column layout:

- Invoice Number (read-only after creation)
- Vendor (dropdown in DRAFT, read-only after submission)
- Invoice Date (date picker in DRAFT, read-only after submission)
- Due Date (date picker in DRAFT, read-only after submission)
- Amount (numeric input in DRAFT, read-only after submission)
- Currency (dropdown in DRAFT, read-only after submission)
- Status (always read-only, auto-managed)
- Created By (always read-only)
- Created At (always read-only)

Line Items Section (Facet 2):

- Table with columns: Line No, Description, Quantity, Unit Price, Total Amount
- In DRAFT status:
 - "Add" button above table (adds new row)
 - "Delete" icon on each row (removes row)
 - All fields editable inline
 - Auto-calculate total amount on quantity/price change
- After DRAFT status:
 - Read-only table, no add/delete buttons
- Footer row: Shows "Total: [SUM]" calculated from all line items
- Warning message if sum $\neq$ header amount (only in DRAFT)

Attachments Section (Facet 3):

- Table with columns: File Name, File Size, Uploaded By, Uploaded At, Actions
- "Upload" button (enabled in DRAFT and SUBMITTED, disabled after)
- File upload accepts: PDF, DOCX, XLSX, PNG, JPG (max 5MB per file)
- "Download" icon on each attachment
- "Delete" icon (only in DRAFT, only for user who uploaded)
- Show file icon based on file type

Approval History Section (Facet 4):

- Timeline view (vertical) showing all approval actions
- Each entry shows:
 - Action type icon (Submitted/Approved/Rejected)
 - Actor name
 - Timestamp (formatted as "DD-MMM-YYYY HH:mm:ss")
 - Comments/Reason
- Sorted chronologically (oldest first, newest last)
- Auto-refresh when new action performed

Actions Bar (Top of page):

- "Edit" button: Visible only in DRAFT status for creator/admin
- "Save" button: Visible in edit mode
- "Cancel" button: Visible in edit mode
- "Submit for Approval" button: Visible in DRAFT status for creator/admin
- "Approve" button: Visible in SUBMITTED status for approver (not submitter)
- "Reject" button: Visible in SUBMITTED status for approver
- "Back" button: Always visible, navigates to list page

Vendor List Page (Admin View)

Initial Load:

- Display all vendors in a table with columns:
 - Vendor Name (clickable link)

- Email
- Phone
- Country
- Currency
- Tax ID
- Status (with colored badge)
- Default sorting: Vendor Name ascending (A-Z)

Status Values:

- PENDING: Newly synced or created, not yet approved
- APPROVED: Can be used for invoice creation
- SUSPENDED: Cannot be used for new invoices
- DELETED: Soft-deleted, hidden from normal views

Filter Bar:

- Status dropdown (multi-select)
- Country dropdown (multi-select)
- Currency dropdown (multi-select)
- Search box for vendor name/email

Actions:

- "Create" button → Opens vendor creation form
- "Sync from S/4HANA" button (Admin only) → Triggers sync action
- Click vendor name → Opens vendor detail page

Analytics/Reports Page

Dashboard Layout:

KPI Cards (Top Row):

1. **Total Invoices**
 - Number: Count of all invoices
 - Trend: % change from previous period
2. **Total Amount**
 - Number: Sum of all invoice amounts
 - Currency: Default currency with conversion note
3. **Pending Approvals**
 - Number: Count of SUBMITTED status invoices
 - Clickable: Links to filtered list
4. **Rejected Invoices**
 - Number: Count of REJECTED status invoices
 - Clickable: Links to filtered list

Charts (Middle Section):

1. **Invoice Status Breakdown (Pie Chart):**
 - Segments: DRAFT, SUBMITTED, APPROVED, REJECTED
 - Show count and percentage for each
 - Click segment → Filters invoice list
2. **Invoice Trend (Line Chart):**
 - X-axis: Last 12 months
 - Y-axis: Invoice count
 - Line: Number of invoices created per month
 - Hover: Show exact count
3. **Top 5 Vendors by Invoice Amount (Bar Chart):**
 - X-axis: Vendor names
 - Y-axis: Total invoice amount
 - Bars: Colored by vendor
 - Click bar → Navigate to vendor detail

Data Table (Bottom Section):

- Aggregated data by vendor showing:
 - Vendor Name
 - Total Invoices
 - Draft Count
 - Submitted Count
 - Approved Count
 - Rejected Count
 - Total Amount
 - Currency
 - Sortable by any column
 - Export to Excel button
-

Row-Level Security Behavior

Vendor Manager User

Login: User logs in with VendorManager role

Vendor List:

- Sees ONLY vendors where vendor.assignedManager = logged-in user ID
- Cannot see other vendors
- Cannot access admin functions

Invoice List:

- Sees ONLY invoices where invoice.vendor.assignedManager = logged-in user ID
- Cannot see invoices for other vendors
- "Create" button enabled (can create invoices only for their vendors)

Vendor Dropdown (in invoice creation):

- Shows ONLY vendors assigned to this manager
- Other vendors not visible in dropdown

Invoice Actions:

- Can edit/submit ONLY invoices for their assigned vendors
- Cannot approve/reject (no Approver role)

Analytics:

- Shows data ONLY for their assigned vendors
- KPIs and charts reflect filtered data

Error on Unauthorized Access:

- If user tries to access invoice URL for another vendor:
 - Error: "Access denied. You do not have permission to view this invoice."
 - HTTP 403 Forbidden

Work Zone Integration Behavior

Launchpad Page Load

User Logs In:

1. Redirect to Work Zone site URL
2. Authenticate via XSUAA (SSO)
3. Load launchpad page based on user role

Tile Visibility by Role:

Tile	Admin	VendorManager	Approver	Viewer
Manage Invoices	✓	✓	X	X
Manage Vendors	✓	X	X	X
Approve Invoices	✓	X	✓	X

Invoice Reports	✓	✓	✓	✓
Sync from S/4HANA	✓	✗	✗	✗

Tile Click Behavior:

- Click "Manage Invoices" → Opens invoice list page
- Click "Manage Vendors" → Opens vendor list page
- Click "Approve Invoices" → Opens invoice list filtered by status=SUBMITTED
- Click "Invoice Reports" → Opens analytics dashboard
- Click "Sync from S/4HANA" → Triggers sync action, shows progress dialog

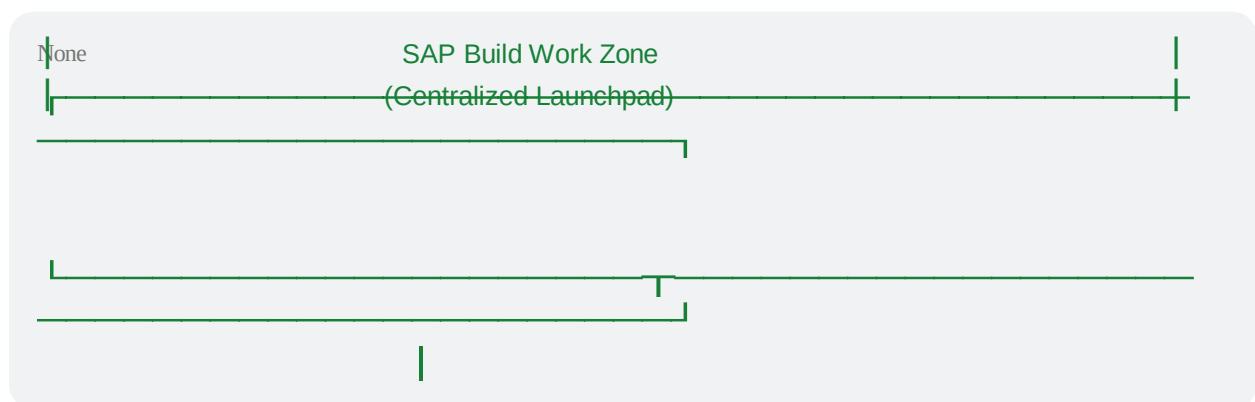
Search in Work Zone:

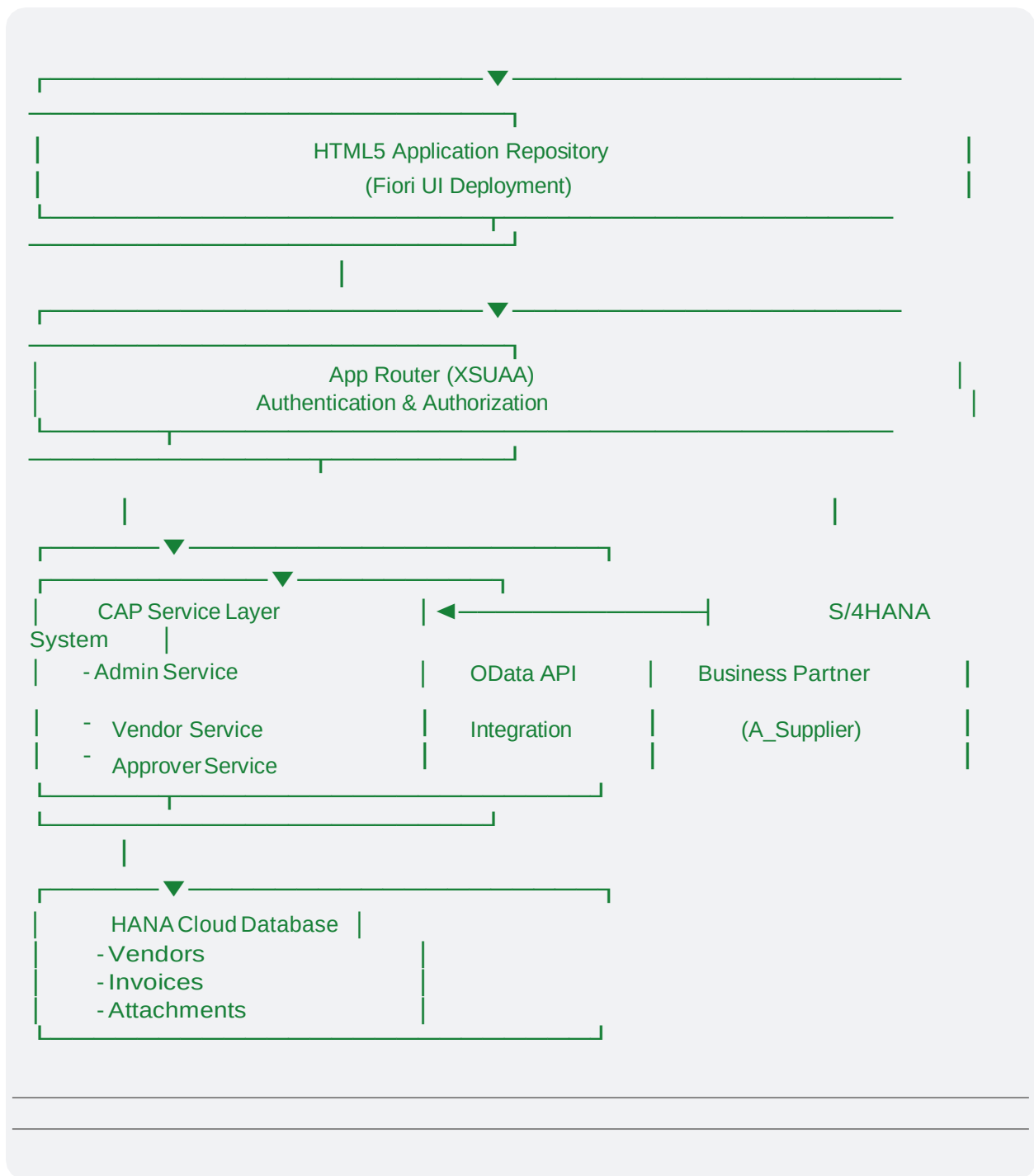
- User types in global search box
- Search across all invoices (respecting role-based access)
- Results show:
 - Invoice number
 - Vendor name
 - Amount
 - Status
- Click result → Navigate to invoice detail page

Notifications (If Implemented):

- Approver sees notification badge when new invoices submitted
- Creator sees notification when invoice approved/rejected
- Click notification → Navigate to relevant invoice

Technical Architecture





Day 42: Planning & Data Model Design

Objectives

- Analyze requirements and design solution architecture
- Create comprehensive CDS data model
- Implement sample data and verify deployment

Expected Outputs & Behavior

Database Schema Verification

After deploying to local database, accessing the service endpoint should show:

Entities Available:

- Vendors
- Invoices
- InvoiceItems
- Attachments
- ApprovalHistory

Sample Data Requirements:

Vendors Table (minimum 10 records):

None

Example records required:

1. Acme Corp - APPROVED - USD - USA
2. Tech Solutions Ltd - APPROVED - EUR - UK
3. Global Supplies - APPROVED - INR - India
4. New Vendor Inc - PENDING - USD - USA
5. Quality Parts - APPROVED - EUR - Germany
6. Supply Chain Co - APPROVED - GBP - UK
7. Materials Plus - REJECTED - USD - Canada
8. Service Provider - APPROVED - AUD - Australia
9. Industrial Goods - PENDING - EUR - France
10. Equipment Rental - APPROVED - USD - USA

Invoices Table (minimum 20 records showing all states):

None

Required state distribution:

- 5 DRAFT invoices (from Admin and VendorManager users)
- 5 SUBMITTED invoices (ready for approval)
- 6 APPROVED invoices (showing successful workflow)
- 4 REJECTED invoices (with rejection reasons)

Example record:

```
INV-001 | Acme Corp | 2024-12-01 | 50,000 | USD | APPROVED
INV-002 | Tech Solutions | 2024-12-05 | 25,000 | EUR | SUBMITTED
INV-003 | Global Supplies | 2024-12-10 | 15,000 | INR | REJECTED
...
```

InvoiceItems Table (minimum 4 records):

None

Each invoice must have 2-5 line items

Line items must sum to invoice total

Example for INV-001 (Total: 50,000):

Line 1: "Software Licenses" | Qty: 10 | Price: 3,000 | Total: 30,000

Line 2: "Support Services" | Qty: 4 | Price: 5,000 | Total: 20,000
Sum = 50,000 ✓

ApprovalHistory Table:

None

For each SUBMITTED/APPROVED/REJECTED invoice, there must be history entries:

Example for INV-001:

Entry 1: SUBMITTED | John Doe | 2024-12-01 10:30 | "Submitted for approval"

Entry 2: APPROVED | Jane Smith | 2024-12-01 15:45 | "Approved - all documents verified"

Verification Tests

When you access <http://localhost:4004> after deployment:

Test 1: Entity Accessibility

- Click on Vendors → Should show 10 vendor records
- Click on Invoices → Should show 20 invoice records
- Click on InvoiceItems → Should show 40+ line item records

Test 2: Association Navigation

- Click on an Invoice → Click on "vendor" navigation → Should open vendor detail
- Click on Vendor → Click on "invoices" navigation → Should show all invoices for that vendor

Test 3: Data Integrity

- For each invoice: Sum of line items = Invoice header amount
- For APPROVED invoices: approvedBy and approvedAt must be populated
- For REJECTED invoices: rejectedBy, rejectedAt, and rejectionReason must be populated
- Invoice dates must not be future dates

Requirements

1. Requirements Analysis & Documentation

Document your understanding of the business requirements:

- Create user stories for each role (Admin, Vendor Manager, Approver, Viewer)
- Define the complete invoice lifecycle with state transitions
- Identify all business entities and their relationships
- Create a detailed ER diagram showing all entities, attributes, and relationships

2. Data Model Design

Design and implement a complete database schema that includes:

Required Entities:

- Vendors with complete contact information, tax details, and approval status
- Invoices with header information, amounts, dates, and workflow status
- Invoice line items with item-level details
- Attachments for invoice documents
- Approval history for audit trail

Data Model Requirements:

- Implement proper primary keys and relationships
- Use appropriate data types for all fields
- Include audit fields (created by, created at, modified by, modified at)
- Define enumerations for status fields

- Implement validations at the database level
- Add value help annotations where applicable
- Ensure referential integrity through proper associations and compositions

3. Sample Data

Create comprehensive sample data covering:

- Minimum 10 vendors with varied attributes (different countries, currencies, approval statuses)
- Minimum 20 invoices in different workflow states
- Minimum 40 line items across invoices
- Minimum 10 attachment records
- Approval history entries showing workflow progression

Data Quality Requirements:

- Data must represent realistic business scenarios
- Include edge cases (rejected invoices, high-value invoices, multi-currency)
- Ensure data consistency across related entities
- Include test data for each business rule validation

4. Local Verification

Deploy and verify your data model:

- Successfully deploy to local database
- All entities accessible via service endpoints
- Sample data loaded correctly
- Associations navigable in both directions
- No database errors or warnings

Deliverables (Day 42)

1. **ER Diagram** - Comprehensive diagram showing all entities and relationships (PDF/PNG)
2. **Data Model Files** - Complete CDS schema files
3. **Sample Data** - CSV files for all entities with quality test data
4. **Deployment Evidence** - Screenshots showing successful deployment and data
5. **Documentation** - Brief document explaining design decisions and assumptions

Evaluation Criteria (Day 42) - 30 Points

Criteria

Points

ER diagram clarity, completeness, and professional quality	5
Correct use of CDS features (aspects, associations, compositions, enums)	8
Data type selection, validations, and constraints	5
Sample data quality, variety, and realism	5
Code organization, naming conventions, and documentation	4
Successful local deployment with no errors	3

Day 43: Service Layer & Fiori UI

Objectives

- Build service layer with business logic
- Implement approval workflow actions
- Generate and customize Fiori Elements UI
- Integrate with S/4HANA for vendor master data

Requirements

1. Service Layer Architecture

Design and implement multiple services for different user roles:

Admin Service:

- Full CRUD access to all entities (Vendors, Invoices, InvoiceItems, Attachments, ApprovalHistory)
- Analytics entity showing aggregated invoice data by vendor, status, and currency
- Restricted to Admin role only
- Support for draft functionality on invoices

Vendor Manager Service:

- Read/write access to invoices for assigned vendors only
- Action to submit invoices for approval
- Row-level security ensuring users only see their vendor's data

- Restricted to VendorManager role

Approver Service:

- Read-only access to all submitted invoices
- Action to approve invoices with comments
- Action to reject invoices with mandatory reason
- Restricted to Approver role

Viewer Service:

- Read-only access to approved invoices
- Access to invoice analytics
- Restricted to Viewer role

2. Business Logic Implementation Implement

all business rules and validations:

Invoice Creation & Modification:

- Validate vendor is in APPROVED status before allowing invoice creation
- Validate invoice date is not in the future
- Validate invoice amount is positive and within allowed limit
- Validate invoice number uniqueness per vendor
- Auto-calculate invoice total from line items
- Ensure at least one line item exists before submission

Submit for Approval Action:

- Validate invoice is in DRAFT status
- Validate all mandatory fields are populated
- Validate line items sum matches invoice total
- Change status to SUBMITTED
- Record submission in approval history

Approve Invoice Action:

- Validate invoice is in SUBMITTED status
- Validate approver is not the submitter
- Change status to APPROVED
- Record approver and approval date
- Log approval in history with comments

Reject Invoice Action:

- Validate invoice is in SUBMITTED status
- Require rejection reason (mandatory)
- Change status to REJECTED

- Log rejection in history with reason

3. S/4HANA Integration

Implement integration with S/4HANA Business Partner API:

Requirements:

- Import S/4HANA Business Partner (Supplier) API metadata
- Create a dedicated integration service
- Implement sync action to fetch vendors from S/4HANA
- Map S/4HANA supplier fields to local vendor fields
- Handle upsert logic (update existing, create new)
- Return sync summary (count of created/updated vendors)
- Configure proper destination for production use
- Handle integration errors gracefully
- Restrict sync action to Admin role only

Data to Sync:

- Supplier name, address, email, phone
- Country, currency, tax ID
- Default new synced vendors to PENDING status

4. Fiori Elements UI

Create a professional Fiori Elements application:

List Report Requirements:

- Display invoices with key information (number, vendor, date, amount, status)
- Enable filtering by vendor, status, date range
- Show status with appropriate criticality (colors)
- Support for searching across fields
- Multi-select for batch operations
- Responsive design for mobile access

Object Page Requirements:

- Header showing invoice summary and status
- Multiple facets/sections:
 - General Information (invoice details)
 - Line Items (editable table)
 - Attachments (upload/download capability)
 - Approval History (read-only timeline)
- Actions visible based on invoice status and user role
- Value helps for vendor selection
- Draft functionality for incomplete invoices

- Field validations with meaningful error messages

UI Annotations:

- Proper labels and descriptions for all fields
- Criticality for status visualization
- Value helps for foreign keys
- Field groupings for better organization
- Navigation from invoice to vendor master
- Semantic coloring for amounts and statuses

5. Testing & Validation Thoroughly

test all functionalities:

Workflow Testing:

- Complete invoice creation with multiple line items
- Submit invoice for approval
- Approve invoice as approver
- Reject invoice with reason
- Verify status transitions are correct
- Test draft save and resume functionality

Validation Testing:

- Try creating invoice for unapproved vendor (should fail)
- Try future-dated invoice (should fail)
- Try approving own submitted invoice (should fail)
- Try approving already approved invoice (should fail)
- Test negative and zero amounts (should fail)
- Test line items total mismatch (should fail)

Integration Testing:

- Execute S/4HANA vendor sync
- Verify vendors created/updated in local database
- Verify data mapping accuracy
- Test sync error handling

Role-Based Testing:

- Test as Admin (full access)
- Test as Vendor Manager (restricted access)
- Test as Approver (approval actions only)
- Test as Viewer (read-only)

Expected Test Scenarios & Results

Test Scenario 1: Create Invoice (Success Path)

Steps:

1. Login as Admin
2. Navigate to invoice list
3. Click "Create" button
4. Select vendor "Acme Corp" (APPROVED status)
5. Enter invoice number "INV-TEST-001"
6. Set invoice date = today
7. Set amount = 10,000
8. Set currency = USD
9. Click "Save"
10. Add line item: "Consulting Services" | Qty: 2 | Price: 5,000
11. Click "Save Draft"

Expected Result:

- Invoice created successfully
 - Status = DRAFT
 - Success message displayed
 - Invoice appears in list with grey "DRAFT" badge
 - Line item total (10,000) matches header amount ✓
-

Test Scenario 2: Submit Invoice (Validation Failure)

Steps:

1. Open invoice INV-TEST-001
2. Delete the line item
3. Click "Submit for Approval"

Expected Result:

- Error dialog appears
 - Error message: "Invoice must have at least one line item"
 - Status remains DRAFT
 - Submit action cancelled
 - Line items section highlighted in red
-

Test Scenario 3: Submit Invoice (Success)

Steps:

1. Open invoice INV-TEST-001
2. Ensure line item exists (sum = 10,000)
3. Click "Submit for Approval"

Expected Result:

- Status changes from DRAFT to SUBMITTED
 - Success message: "Invoice INV-TEST-001 submitted for approval successfully"
 - Status badge changes to yellow "SUBMITTED"
 - All fields become read-only
 - Edit button disappears
 - Approve/Reject buttons appear (if viewing as Approver)
 - Approval history shows: "SUBMITTED by [Admin Name] on [Date Time]"
-

Test Scenario 4: Approve Own Invoice (Validation Failure)

Steps:

1. Remain logged in as Admin (who submitted INV-TEST-001)
2. Attempt to click "Approve" button

Expected Result:

- Error dialog appears
 - Error message: "You cannot approve your own submitted invoice"
 - Status remains SUBMITTED
 - Approve action cancelled
-

Test Scenario 5: Approve Invoice (Success)

Steps:

1. Logout from Admin
2. Login as Approver user
3. Navigate to invoices
4. Open INV-TEST-001 (status = SUBMITTED)
5. Click "Approve" button
6. Enter comments: "Invoice approved after document verification"
7. Confirm approval

Expected Result:

- Status changes from SUBMITTED to APPROVED
- Success message: "Invoice INV-TEST-001 approved successfully by [Approver Name]"
- Status badge changes to green "APPROVED"

- Approve/Reject buttons disappear
 - Header shows: "Approved by [Approver Name] on [Date Time]"
 - Approval history shows new entry: "APPROVED by [Approver Name] on [Date Time] - Invoice approved after document verification"
-

Test Scenario 6: Reject Invoice

Steps:

1. Login as Approver
2. Open another invoice with status = SUBMITTED
3. Click "Reject" button
4. Try to confirm without entering reason

Expected Result:

- Validation error: "Rejection reason is mandatory"
- Cannot proceed

Continue Steps: 5. Enter reason: "Missing supporting documents" 6. Confirm rejection

Expected Result:

- Status changes from SUBMITTED to REJECTED
 - Status badge changes to red "REJECTED"
 - Red alert box shows: "Invoice rejected by [Approver Name] - Missing supporting documents"
 - No further actions available
 - Approval history shows: "REJECTED by [Approver Name] on [Date Time] - Missing supporting documents"
-

Test Scenario 7: S/4HANA Vendor Sync

Steps:

1. Login as Admin
2. Click "Sync from S/4HANA" tile in Work Zone
3. Wait for process to complete

Expected Result:

- Loading indicator: "Syncing vendors from S/4HANA..."
- After 3-10 seconds, success dialog:

None

Vendor sync completed successfully!

- Total vendors in S/4HANA: 15
- New vendors created: 3
- Existing vendors updated: 12
- Sync completed at: 25-JUN-2024 14:30:15

- Navigate to vendor list
 - New vendors appear with status = PENDING
 - Existing vendors show updated information
-

Test Scenario 8: Row-Level Security (VendorManager)

Steps:

1. Login as VendorManager user (assigned to vendor "Acme Corp")
2. Navigate to invoice list

Expected Result:

- Only invoices for "Acme Corp" visible
- Invoices for other vendors NOT visible
- Create button enabled

Continue Steps: 3. Click "Create" button 4. Open vendor dropdown

Expected Result:

- Dropdown shows ONLY "Acme Corp"
- Other vendors not visible in dropdown

Continue Steps: 5. Try to access URL of invoice for another vendor (e.g., INV-002 for "Tech Solutions")

Expected Result:

- Error page: "Access denied. You do not have permission to view this invoice."
 - HTTP 403 status
-

Test Scenario 9: Draft Functionality

Steps:

1. Login as Admin

2. Create new invoice
3. Fill in some fields (vendor, date)
4. Do NOT click "Save" - close browser tab
5. Reopen application
6. Check invoice list

Expected Result:

- Incomplete invoice NOT saved (no draft auto-save)

Continue Steps: 7. Create new invoice 8. Fill all required fields 9. Click "Save" (status = DRAFT) 10. Add one line item 11. Close browser (without clicking Submit) 12. Reopen application 13. Find invoice in list

Expected Result:

- Invoice exists with status = DRAFT
- Contains saved line item
- Can be edited and completed

Deliverables (Day 43)

1. **Service Definition Files** - All service CDS and implementation files
2. **S/4HANA Integration** - Complete integration service with sync functionality
3. **Fiori Application** - Fully annotated and functional UI
4. **Test Results Document** - Screenshots showing ALL 9 test scenarios with actual vs expected results
5. **Demo Video** (2-3 minutes) - End-to-end workflow demonstration showing:
 - Create invoice with line items
 - Submit for approval (show validation working)
 - Approve as different user
 - Show rejection flow
 - Demonstrate S/4HANA sync with before/after vendor counts

Evaluation Criteria (Day 43) - 40 Points

Criteria	Points
Service layer design and role-based access control	6
Business logic correctness and completeness	10

S/4HANA integration implementation and error handling	8
Workflow actions implementation (submit, approve, reject)	8
Fiori UI quality, annotations, and user experience	6
Testing thoroughness and documentation	2

Day 44: Security, Deployment & Build Work Zone

Objectives

- Implement enterprise-grade security
- Deploy to Cloud Foundry with HANA Cloud
- Configure SAP Build Work Zone access
- Conduct end-to-end testing in production environment

Requirements

1. Security Implementation

Design and implement comprehensive security:

Authentication & Authorization:

- Define security scopes for all four roles (Admin, VendorManager, Approver, Viewer)
- Create role templates mapping scopes to roles
- Configure XSUAA service instance
- Implement service-level authorization restrictions
- Implement row-level security for Vendor Managers

Role Definitions:

- **Admin:** Full access to all entities and operations
- **VendorManager:** Access only to invoices for assigned vendors
- **Approver:** Read access to submitted invoices, approval/rejection actions
- **Viewer:** Read-only access to approved invoices and reports

Security Requirements:

- Row-level filtering must prevent unauthorized data access
- Actions must be protected by role checks
- Business logic must validate user permissions
- Audit fields must capture user identity

2. Multi-Target Application (MTA) Configuration

Create complete MTA descriptor for Cloud Foundry deployment:

Required Modules:

- CAP service module (Node.js runtime)
- HANA database deployer module
- HTML5 application deployer module
- Destination configuration module

Required Resources:

- HANA Cloud HDI container
- XSUAA service instance with security descriptor
- Destination service instance
- HTML5 Application Repository host service
- HTML5 Application Repository runtime service

Destination Configuration:

- Configure S/4HANA system destination
- Set appropriate authentication method
- Configure proxy settings if required
- Include all necessary parameters

Build Parameters:

- Configure build packs
- Define module dependencies
- Set environment variables
- Configure service bindings

3. Cloud Foundry Deployment

Deploy the complete application to Cloud Foundry:

Pre-Deployment:

- Ensure HANA Cloud instance is created and running
- Map HANA Cloud to your Cloud Foundry space

- Verify Cloud Foundry CLI is configured
- Build MTA archive successfully

Deployment Tasks:

- Deploy MTA archive to Cloud Foundry
- Verify all applications are in running state
- Verify all services are bound correctly
- Test service endpoints are accessible
- Verify data is deployed to HANA Cloud

Post-Deployment Verification:

- Check application logs for errors
- Test database connectivity
- Verify XSUAA integration
- Test S/4HANA destination connectivity

4. Role Collection Assignment Configure

BTP security for user access:

Role Collection Creation:

- Create role collections for each role template
- Map role templates to role collections
- Document role collection purposes

User Assignment:

- Assign yourself to Admin role collection
- Create/assign test users to other role collections
- Verify role assignments are active
- Document user-to-role mappings

5. SAP Build Work Zone Configuration Configure

Work Zone as the central entry point:

Site Setup:

- Subscribe to SAP Build Work Zone (if not already)
- Create new site for Vendor Invoice Management
- Configure site branding and theme
- Set up content providers for HTML5 apps

Launchpad Page Design: Create an organized launchpad with tiles for:

- Invoice management (Admin)

- Vendor management (Admin)
- Invoice approval (Approver)
- Invoice reports/analytics (All roles)
- S/4HANA vendor sync (Admin)

Tile Configuration:

- Configure each tile with proper navigation
- Set tile visibility based on roles
- Add meaningful tile titles and subtitles
- Configure tile icons appropriately

Advanced Features:

- Enable site search functionality
- Configure search connectors for invoices
- Set up groups for logical organization
- Configure default page for different roles

6. End-to-End Production Testing

Conduct comprehensive testing in the deployed environment:

Workflow Testing:

- Login to Work Zone with different user accounts
- Test complete invoice creation to approval workflow
- Verify S/4HANA vendor sync functionality
- Test draft functionality in cloud
- Verify attachments upload/download
- Test approval history tracking

Security Testing:

- Verify Admin can access all features
- Verify VendorManager sees only assigned vendor invoices
- Verify Approver can approve/reject but not create
- Verify Viewer has read-only access
- Test unauthorized access attempts
- Verify row-level security works correctly

Integration Testing:

- Execute S/4HANA vendor sync from Work Zone
- Verify data synchronization accuracy
- Test error handling for integration failures
- Verify destination configuration works

Performance Testing:

- Test with multiple concurrent users
- Verify response times are acceptable
- Check for memory leaks or resource issues
- Monitor application logs

User Experience Testing:

- Navigate through all Work Zone tiles
- Test on mobile browser
- Verify all UI elements render correctly
- Test search functionality
- Verify all buttons and actions work

Expected Deployment Results

Cloud Foundry Apps (cf apps output)

After successful deployment, **cf apps** command must show:

None				
name	state	instances	memory	
disk				
vendor-invoice-srv	started	1/1	1G	1G
vendor-invoice-app-deployer	stopped	0/1	256M	
256M				
vendor-invoice-db-deployer	stopped	0/1	256M	
256M				

Expected:

- Service app: RUNNING (green)
- Deployer apps: STOPPED (normal after deployment)
- No crashed apps

Test Service URL:

- Click on vendor-invoice-srv URL
- Should open CAP service page
- Should show all service endpoints
- Click on any entity → Should show data from HANA Cloud

Cloud Foundry Services (cf services output)

None

name	service	plan
bound apps		
vendor-invoice-db	hana	hdi-shared
vendor-invoice-srv		
vendor-invoice-uaa	xsuaa	application
vendor-invoice-srv		
vendor-invoice-destination	destination	lite
vendor-invoice-srv		
vendor-invoice-html5-repo-host	html5-apps-repo	app-host
vendor-invoice-app-deployer		
vendor-invoice-html5-repo-rt	html5-apps-repo	app-runtime
vendor-invoice-srv		

Expected:

- All services in "create succeeded" status
- All services bound to respective apps
- No service errors

Work Zone Expected Behavior Test

Case 1: Admin User Login Steps:

1. Open Work Zone URL
2. Login with Admin user credentials
3. Observe launchpad

Expected Result:

None

Vendor Invoice Management		
[Manage Invoices]	[Manage Vendors]	
[Approve Invoices]	[Invoice Reports]	



[Sync from S/4]

- All 5 tiles visible
- Each tile has appropriate icon
- Tiles are clickable

Click "Manage Invoices":

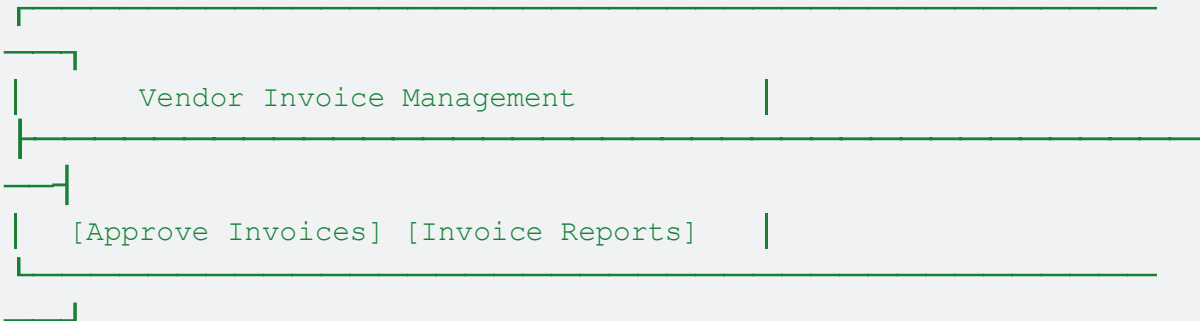
- Opens Fiori app in same window/new tab
- Shows invoice list with data from HANA Cloud
- All CRUD operations work
- No authentication errors

Test Case 2: Approver User Login Steps:

1. Open Work Zone URL
2. Login with Approver user credentials
3. Observe launchpad

Expected Result:

None



Vendor Invoice Management

[Approve Invoices] [Invoice Reports]

- Only 2 tiles visible (Approve Invoices, Reports)
- Manage Invoices, Manage Vendors, Sync tiles NOT visible
- Role-based tile visibility working

Click "Approve Invoices":

- Opens invoice list filtered to SUBMITTED status
- Can open invoice details
- Approve and Reject buttons visible

- Create/Edit buttons NOT visible

Test Case 3: VendorManager User Login Steps:

1. Login as VendorManager (assigned to vendor "Acme Corp")
2. Click "Manage Invoices" tile

Expected Result:

- Invoice list shows ONLY invoices for "Acme Corp"
- 10 invoices for other vendors NOT visible
- Can create new invoice
- Vendor dropdown shows only "Acme Corp"

Test Case 4: Work Zone Search

Steps:

1. Login as Admin
2. Type "INV-001" in Work Zone search bar
3. Press Enter

Expected Result:

- Search results show invoice INV-001
- Result displays: Invoice Number, Vendor, Amount, Status
- Click result → Opens invoice detail page
- Search respects user's role (VendorManager sees only their invoices)

End-to-End Production Workflow Test

Complete flow that must work in deployed environment: Step

1: Admin logs into Work Zone

- **Expected:** Sees all 5 tiles

Step 2: Click "Sync from S/4HANA"

- **Expected:** Progress dialog, then success message with counts

Step 3: Click "Manage Vendors"

- **Expected:** Vendor list with newly synced vendors (status = PENDING)

Step 4: Admin approves a pending vendor

- **Expected:** Status changes to APPROVED

Step 5: Click "Manage Invoices" → Create new invoice

- **Expected:** New approved vendor appears in dropdown

Step 6: Create invoice with 2 line items

- **Expected:** Draft saved successfully

Step 7: Submit invoice

- **Expected:** Status = SUBMITTED, success message

Step 8: Logout, login as Approver

- **Expected:** Sees 2 tiles (not all 5)

Step 9: Click "Approve Invoices"

- **Expected:** Sees submitted invoice in list

Step 10: Open invoice, click Approve

- **Expected:** Approval succeeds, status = APPROVED

Step 11: Click "Invoice Reports"

- **Expected:** Dashboard shows updated counts (1 more approved invoice)

Step 12: Logout, login as VendorManager

- **Expected:** Sees only 2 tiles

Step 13: Click "Manage Invoices"

- **Expected:** Sees ONLY invoices for assigned vendor (not the one just created by Admin for different vendor)

This complete flow proves:

- ✓Deployment successful
 - ✓S/4HANA integration working
 - ✓Role-based access working
 - ✓Workflow actions working
 - ✓Row-level security working
 - ✓Work Zone integration working
-

Deliverables (Day 44)

1. **Security Configuration** - xs-security.json with complete role model
2. **MTA Descriptor** - mta.yaml with all modules, resources, and configurations
3. **Deployment Evidence** - Screenshots showing:
 - cf appsoutput (all apps running)
 - cf servicesoutput (all services created)
 - Service URL with metadata page
 - HANA Cloud instance status
4. **Work Zone Configuration** - Screenshots of:
 - Site manager showing configured site
 - Launchpad for Admin user (5 tiles)
 - Launchpad for Approver user (2 tiles)
 - Each tile's configuration page
5. **Role Assignment Document** - Table format:

User Email	Role Collection	Access Level
admin@company.com	VendorInvoice_Admin	Full access
approver@company.co m	VendorInvoice_Approver	Approve only
manager@company.co m	VendorInvoice_VendorManager	Own vendors

6. **Test Results** - Screenshots showing complete end-to-end flow (Steps 1-13 above)
7. **Architecture Diagram** - Visual showing: Browser → Work Zone → App Router → CAP Service → HANA Cloud, with S/4HANA connection
8. **Final Demo Video** (5 minutes minimum) demonstrating:
 - Login to Work Zone as Admin (show all tiles)
 - Execute S/4HANA sync (show progress and results)
 - Create invoice with line items (show form, validation)
 - Submit for approval (show status change)
 - Logout and login as Approver (show different tiles)
 - Approve the invoice (show approval dialog and history)
 - Show analytics dashboard with updated data
 - Login as VendorManager (show row-level security in action)
 - Use Work Zone search to find invoice
 - Narrate each step clearly explaining what's happening

Evaluation Criteria (Day 44) - 30 Points

Criteria	Points
Security implementation completeness and correctness	6
MTA configuration accuracy and completeness	5
Successful Cloud Foundry deployment (all apps running)	5
HANA Cloud deployment with data migration	4
Work Zone professional configuration and navigation	6
End-to-end testing thoroughness and demo quality	4

Submission Guidelines

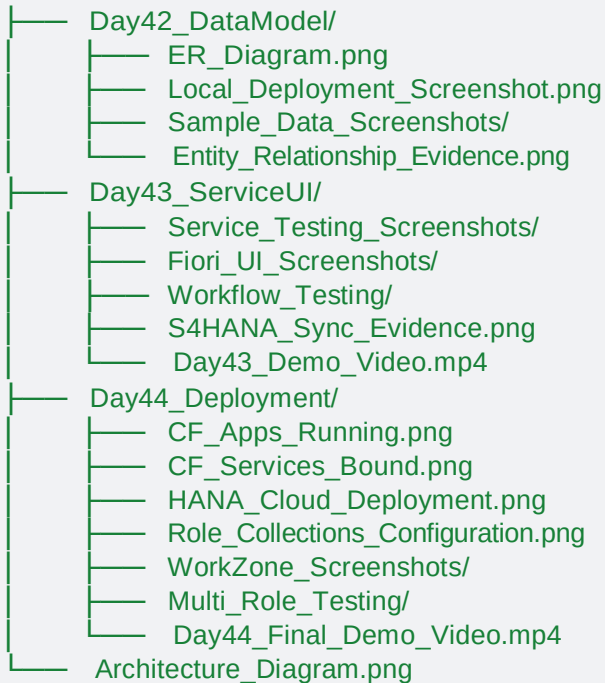
What to Submit

Submit all deliverables via the designated submission portal by the deadline.

1. **Source Code** (GitHub repository link)
 - Complete project source code
 - All CDS models, service implementations, and UI file
 - Sample data files
 - Configuration files (mta.yaml, xs-security.json, package.json)
 - README.md with project overview and setup instructions
 - .gitignore configured properly
2. **Documentation Package** (Single PDF, max 20 pages)
 - **Cover Page:** Project title, your name, student ID, date
 - [Table of Contents](#)
 - **ER Diagram:** Complete entity-relationship diagram
 - **Architecture Diagram:** Deployed solution architecture
 - **Security Model:** Roles and permissions matrix
 - **Service Documentation:** List of services and their purposes
 - **Design Decisions:** Key technical decisions and rationale
 - **Challenges & Solutions:** Problems faced and how you solved them
 - **Test Results Summary:** Overview of testing performed
3. **Evidence Package** (Organized folder structure)

None

Evidence/



4.

Demo Videos (MP4 format)

- **Day 43 Demo** (2-3 minutes): Local workflow demonstration
- **Day 44 Final Demo** (5 minutes): Complete deployed solution demonstration
- Videos should have clear audio narration explaining what you're demonstrating
- Show end-to-end workflows, not just clicking through screens
- Include testing of different user roles

Quality Standards

Your submission will be evaluated on:

- **Code Quality:** Clean, well-organized, properly commented
- **Functionality:** All requirements implemented and working
- **Documentation:** Clear, professional, comprehensive
- **Evidence:** Screenshots clearly show required functionality
- **Videos:** Professional presentation with clear narration